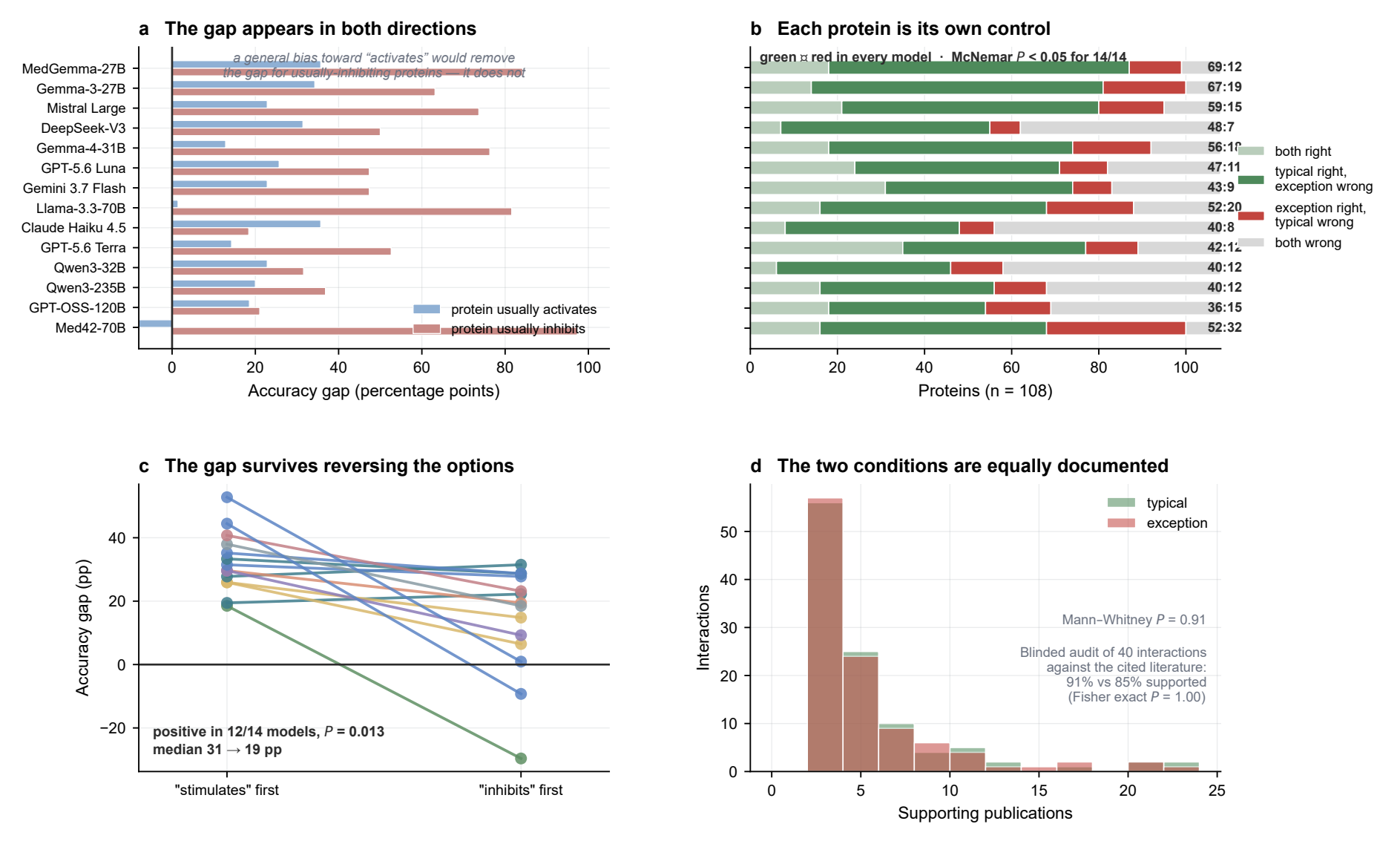

a The gap appears in both directions
a general bias toward "activates" would remove the gap for usually-inhibiting proteins — it does not
MedGemma-27B
Gemma-3-27B
Mistral Large
DeepSeek-V3
Gemma-4-31B
GPT-5.6 Luna
Gemini 3.7 Flash
Llama-3.3-70B
Claude Haiku 4.5
GPT-5.6 Terra
Qwen3-32B
Qwen3-235B
GPT-OSS-120B
Med42-70B
protein usually activates
protein usually inhibits
0
20
40
60
80
100
Accuracy gap (percentage points)
b Each protein is its own control
green ¤ red in every model · McNemar P < 0.05 for 14/14
69:12
67:19
59:15
48:7
56:10
47:11
43:9
52:20
40:8
42:12
40:12
40:12
36:15
52:32
both right
typical right, exception wrong
exception right, typical wrong
both wrong
0
20
40
60
80
100
Proteins (n = 108)
c The gap survives reversing the options
Accuracy gap (pp)
40
20
0
−20
positive in 12/14 models, P = 0.013
median 31 → 19 pp
"stimulates" first
"inhibits" first
d The two conditions are equally documented
typical
exception
Interactions
50
40
30
20
10
0
Mann–Whitney P = 0.91
Blinded audit of 40 interactions against the cited literature: 91% vs 85% supported (Fisher exact P = 1.00)
0
5
10
15
20
25
Supporting publications